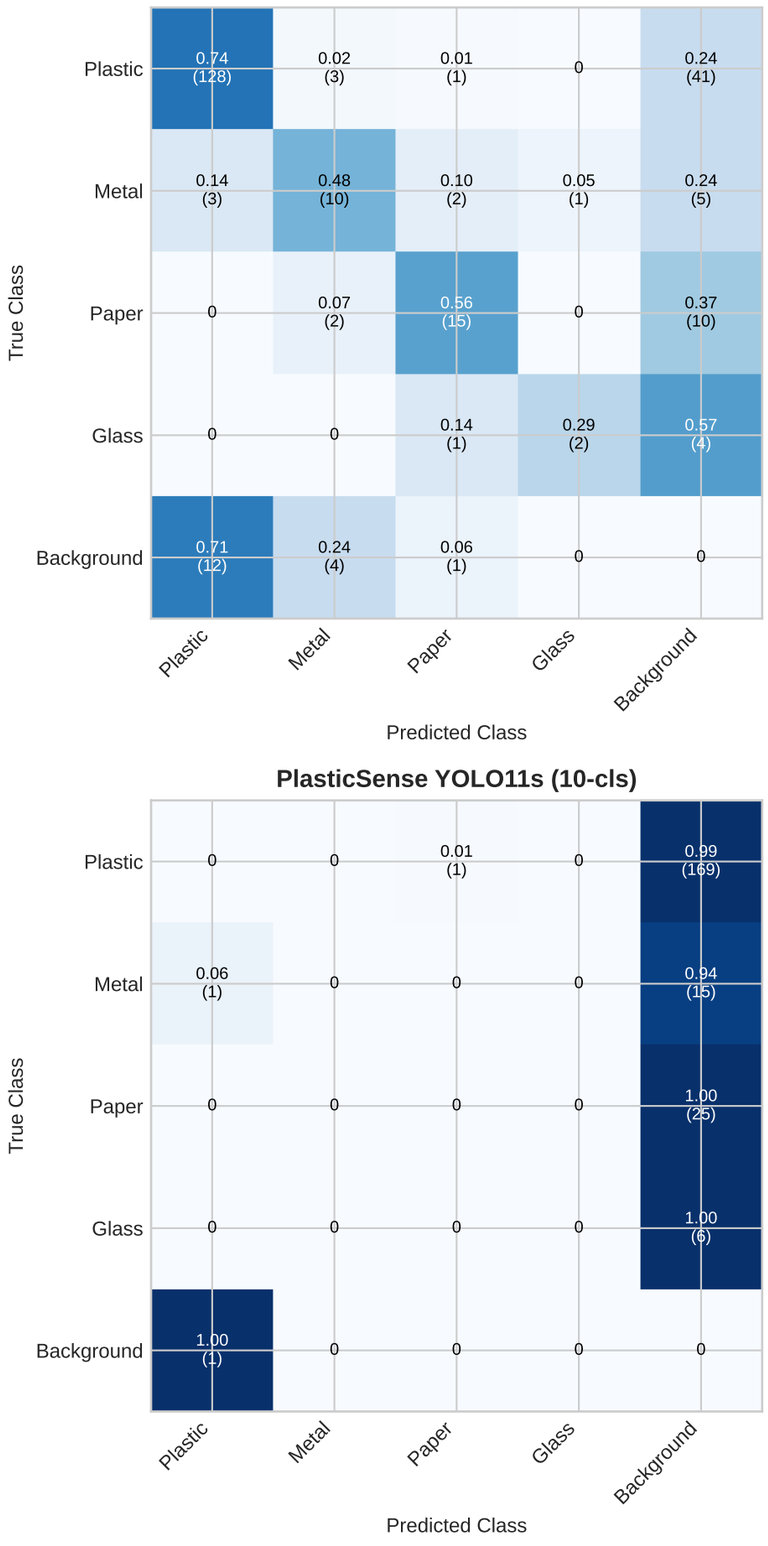
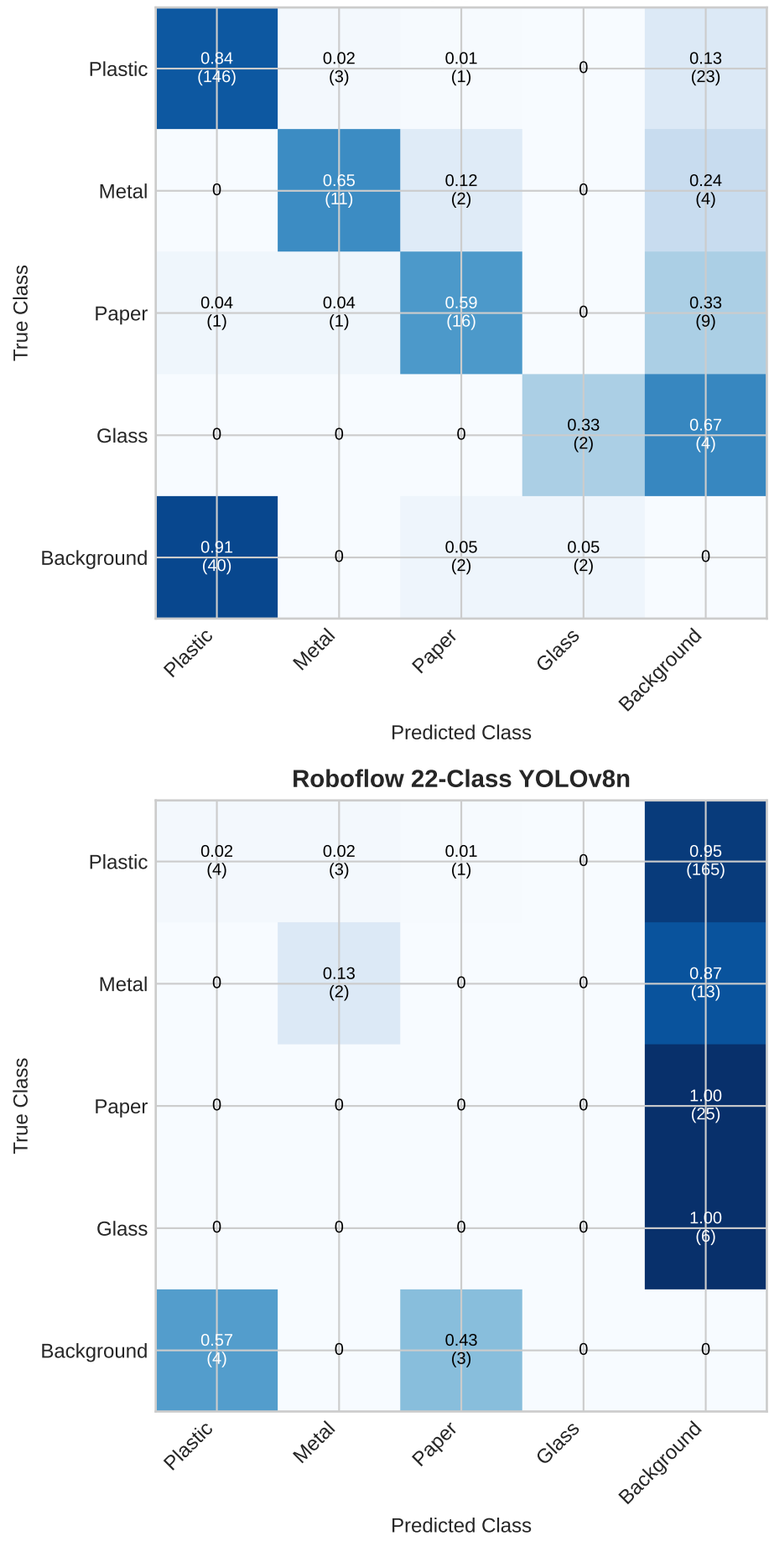
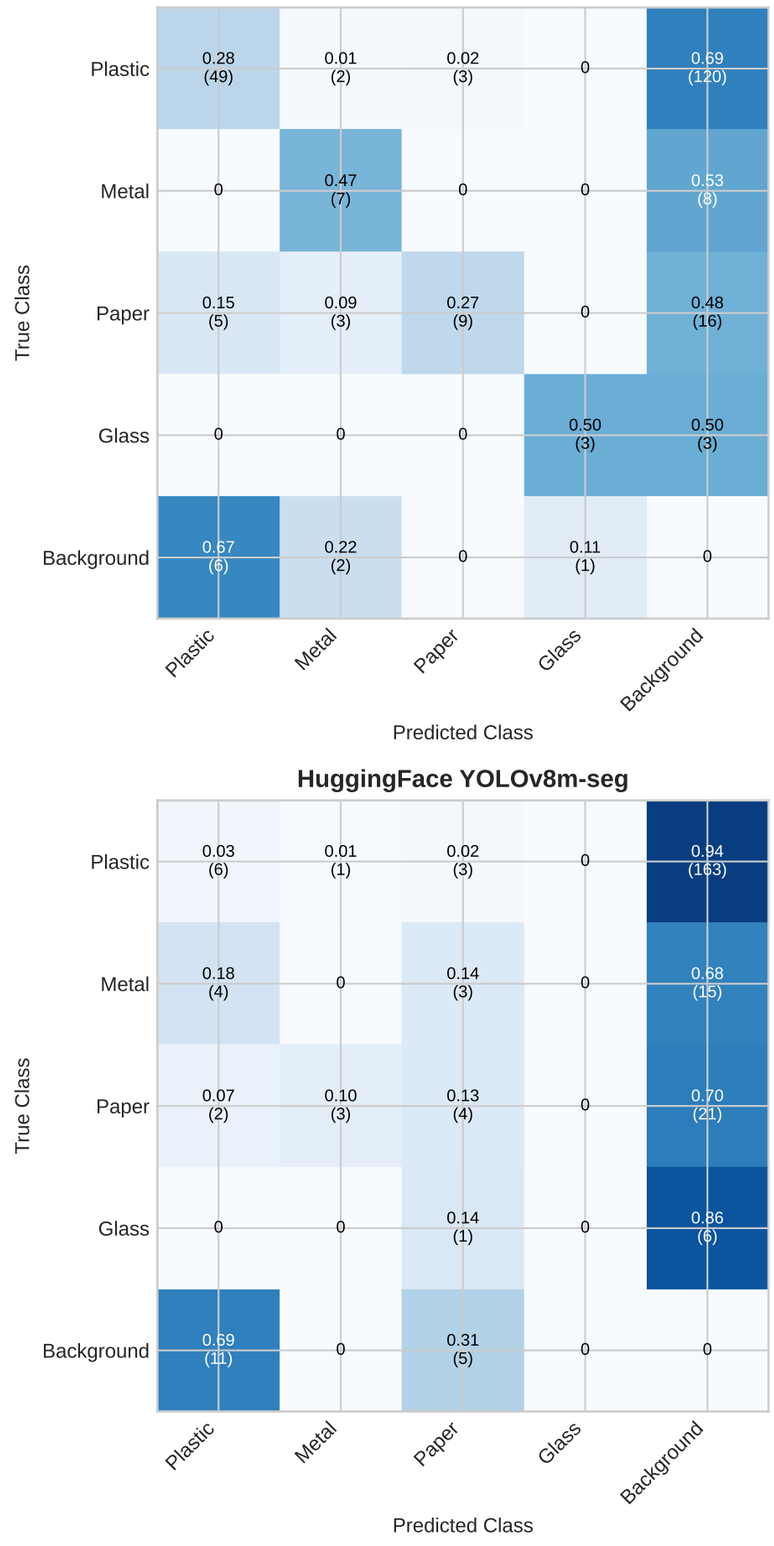

BiaEye YOLO11s (Baseline)
True Class
Predicted Class
Plastic
Metal
Paper
Glass
Background
0.74 (128)
0.02 (3)
0.01 (1)
0
0.24 (41)
0.14 (3)
0.48 (10)
0.10 (2)
0.05 (1)
0.24 (5)
0
0.07 (2)
0.56 (15)
0
0.37 (10)
0
0
0.14 (1)
0.29 (2)
0.57 (4)
0.71 (12)
0.24 (4)
0.06 (1)
0
0


BiaEye YOLO11s (High-Recall + Cleaned)
True Class
Predicted Class
Plastic
Metal
Paper
Glass
Background
0.84 (146)
0.02 (3)
0.01 (1)
0
0.13 (23)
0
0.65 (11)
0.12 (2)
0
0.24 (4)
0.04 (1)
0.04 (1)
0.59 (16)
0
0.33 (9)
0
0
0
0.33 (2)
0.67 (4)
0.91 (40)
0
0.05 (2)
0.05 (2)
0


TACO YOLO11s-seg
True Class
Predicted Class
Plastic
Metal
Paper
Glass
Background
0.28 (49)
0.01 (2)
0.02 (3)
0
0.69 (120)
0
0.47 (7)
0
0
0.53 (8)
0.15 (5)
0.09 (3)
0.27 (9)
0
0.48 (16)
0
0
0
0.50 (3)
0.50 (3)
0.67 (6)
0.22 (2)
0
0.11 (1)
0

PlasticSense YOLO11s (10-cls)
True Class
Predicted Class
Plastic
Metal
Paper
Glass
Background
0
0
0.01 (1)
0
0.99 (169)
0.06 (1)
0
0
0
0.94 (15)
0
0
0
0
1.00 (25)
0
0
0
0
1.00 (6)
1.00 (1)
0
0
0
0

Roboflow 22-Class YOLOv8n
True Class
Predicted Class
Plastic
Metal
Paper
Glass
Background
0.02 (4)
0.02 (3)
0.01 (1)
0
0.95 (165)
0
0.13 (2)
0
0
0.87 (13)
0
0
0
0
1.00 (25)
0
0
0
0
1.00 (6)
0.57 (4)
0
0.43 (3)
0
0

HuggingFace YOLOv8m-seg
True Class
Predicted Class
Plastic
Metal
Paper
Glass
Background
0.03 (6)
0.01 (1)
0.02 (3)
0
0.94 (163)
0.18 (4)
0
0.14 (3)
0
0.68 (15)
0.07 (2)
0.10 (3)
0.13 (4)
0
0.70 (21)
0
0
0.14 (1)
0
0.86 (6)
0.69 (11)
0
0.31 (5)
0
0